Accelerating cosmology

Handley Lab: KICC Welcome Day

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20th October 2026

Accelerating cosmology from theory to data

Handley Lab: Science@KICC <wh260@cam.ac.uk>

- **Big problems**: cosmic acceleration – Λ, evolving dark energy or modified gravity? H_0, S_8 and DESI–SNe tensions: physics, systematics or priors?
- **GPUs for science**: classical Bayes, differential equations and forward models in `JAX` / `BlackJAX`.
- **AI for scientists**: agents for legacy code and algebra; people choose the physics and check the results.
- **The programmes**: dark energy and fundamental theory; real-time gravitational waves; LSST, Euclid and JWST readiness.



The Handley Lab (2026–2027)

Postdocs and fellows


Natalie Hogg


Matt Grayling


Chris Lovell


James Alvey

NS
Nikhil Sarin

PhD students


Dily Ong


Namu Kroupa


Sam Leeney


Toby Lovick


Lawrence Berry


Charlotte Priestley


Farah Wallauer

SV
Selma Vangstein

SH
Sam Hewson


Metha Prathaban


Sinah Legner


Wei-Ning Deng

Part III / MASt 2026-27

JL
Jaiman Lad

JGF
Jesús Girones Fraile

DP
David Parodi

MK
Manaal Khan

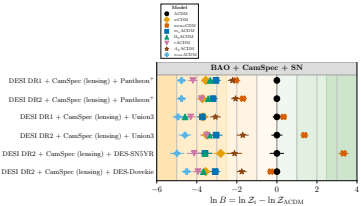
CM
Charlotte McCarthy

AB
Ashwin Balakrishnan

BTB
Benedict Thomas Binu

Dark energy, tensions and fundamental theory to data

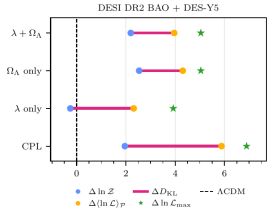
Candidate Lagrangians, GPU Boltzmann solvers and DESI DR2



DESI: evolving dark energy?
Dily Ong, David Yallup, Adam Ormondroyd

Separate evolving dark energy from priors and supernova calibration. [2603.05472]

`unimpeded`: compare models and datasets using public chains. [2511.04661]

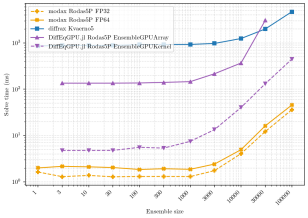


Sinah Legner: thesis, torsion models vs DESI

Fundamental actions to data
Sinah Legner, Wei-Ning Deng, Selma Vangstein, Sam Hewson

Torsion Condensation and CPT-symmetric cosmology. [2507.09228] [2509.10379]

Next: agent-derived perturbations with Cavendish Gravity.



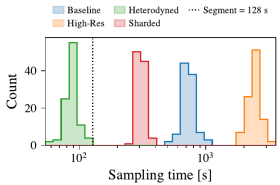
GPU Boltzmann solvers
Lawrence Berry, Oliver Hahn, Nils Schöneberg

`modax`: GPU differential equations. `DISCO-EB`: CMB spectra without tight-coupling approximations. [2609.35661]

Next: evolve candidate theories and compare with data.

Real-time GWs and Stage-IV survey readiness

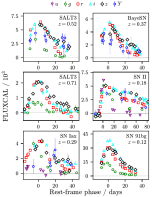
Classical Bayes for LVK, Einstein Telescope, LISA, LSST, Euclid and JWST



Real-time gravitational waves
Metha Prathaban, David Yallup, James Alvey, Nikhil Sarin, Farah Wallauer

GPU likelihoods and `SwiG` nested sampling for binary neutron-star inference. [2607.28265]

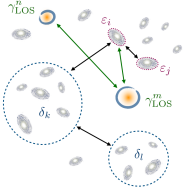
Next: Einstein Telescope, LISA, overlapping signals and shadow sirens.



Time-domain and survey inference
Toby Lovick, Matt Grayling, Namu Kroupa; Sam Leeney, Charlotte Priestley

ALCS: marginalise latent structure. [2603.26644] `JAX-SNANA`: simulate supernovae. [2610.19000]

Quasar delays [2602.22307]; radio transients; satellite orbits.



Lensing, simulations and decisions
Natalie Hogg, Charlotte McCarthy, Chris Lovell; James Alvey, Nikhil Sarin

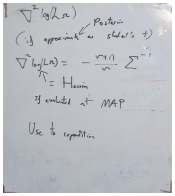
Strong and weak lensing for Euclid / JWST; CAMELS / FLAMELS galaxy simulations.

Next: `BayesClaw` / `LSSTClaw` / `IGWNClaw` for transient tagging, model selection and re-fitting.

The Ambient AI Lab: AI for scientists, by scientists

Cambridge-owned compute, live room loops and how to join in





Local models and scientific tools
Local models and transcription on `turing` / `newton`; Cambridge UIS compute. `Alan` / `Steve`: agent tools using our shared scientific context.

Training without deskilling: students build physical intuition; senior researchers supervise agents.

Natalie Hogg: *Agentic research is oxymoronic*, Nature Astronomy. [2608.31161]

Live room loops
Toby Lovick, James Alvey, Chris Lovell, Natalie Hogg
Whiteboard capture, local transcription and equation retrieval; shared supervision context.

This term: `ai@cam` supervision hardware; Cosmos projects on ambient capture and anti-sycophancy.

People select questions, test the physics and take responsibility.

Join us: Tuesday group meeting, 14:00; Friday arXiv Scrolling Club; Friday Cavendish Gravity, 15:00 (F22); weekly Transients & GWs. `handley-lab.co.uk`